

How Stable Are Regression Conclusions Across Subpopulations?

A FINITE-SAMPLE EMPIRICAL STUDY

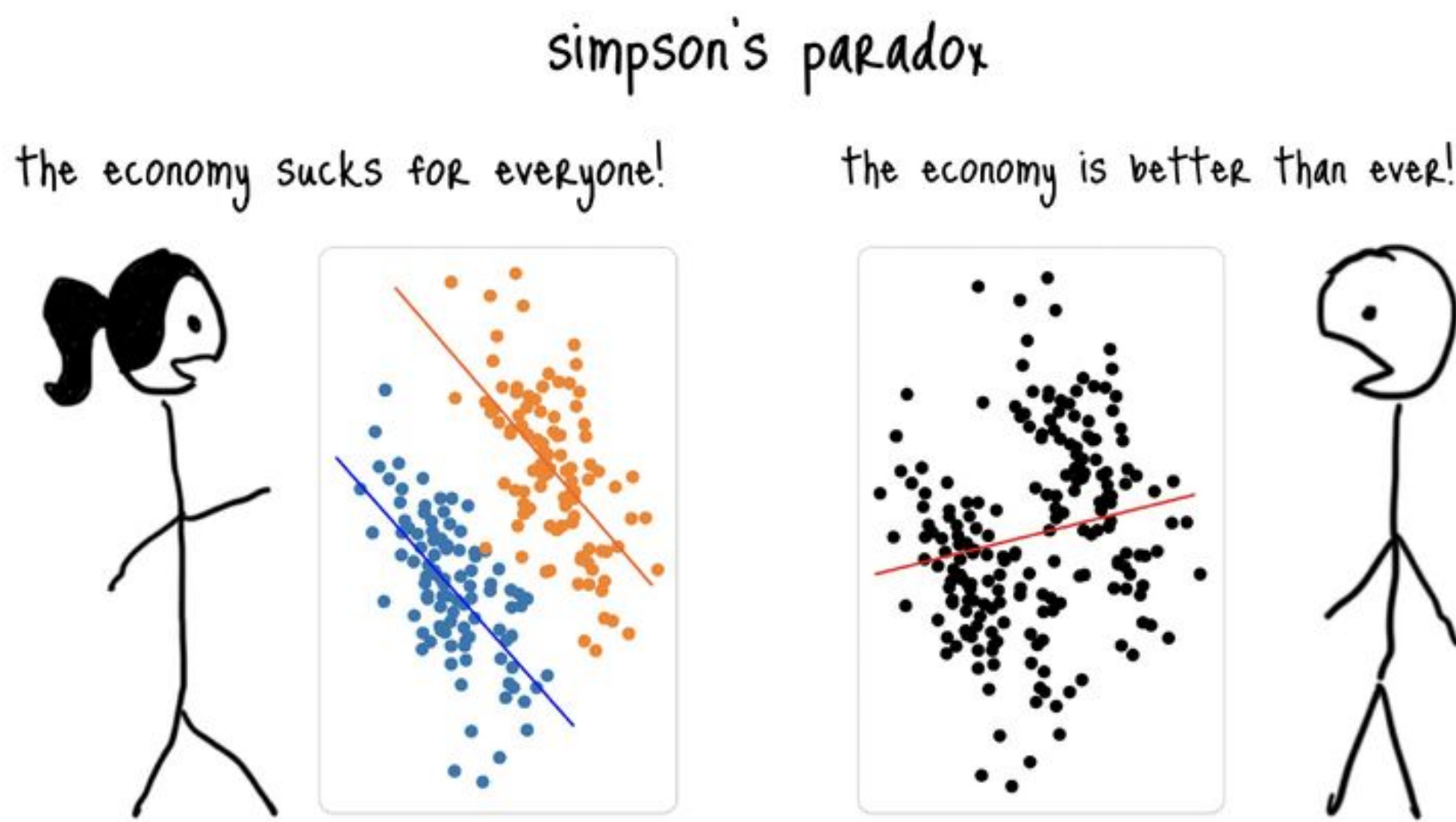
[Authors] – Department of Statistics – July 2026

The Problem

You fit a regression on the full dataset. R^2 is fine, p-values are small. A reviewer asks: “*Does hold for each subgroup?*”

You check. It doesn’t. Now what?

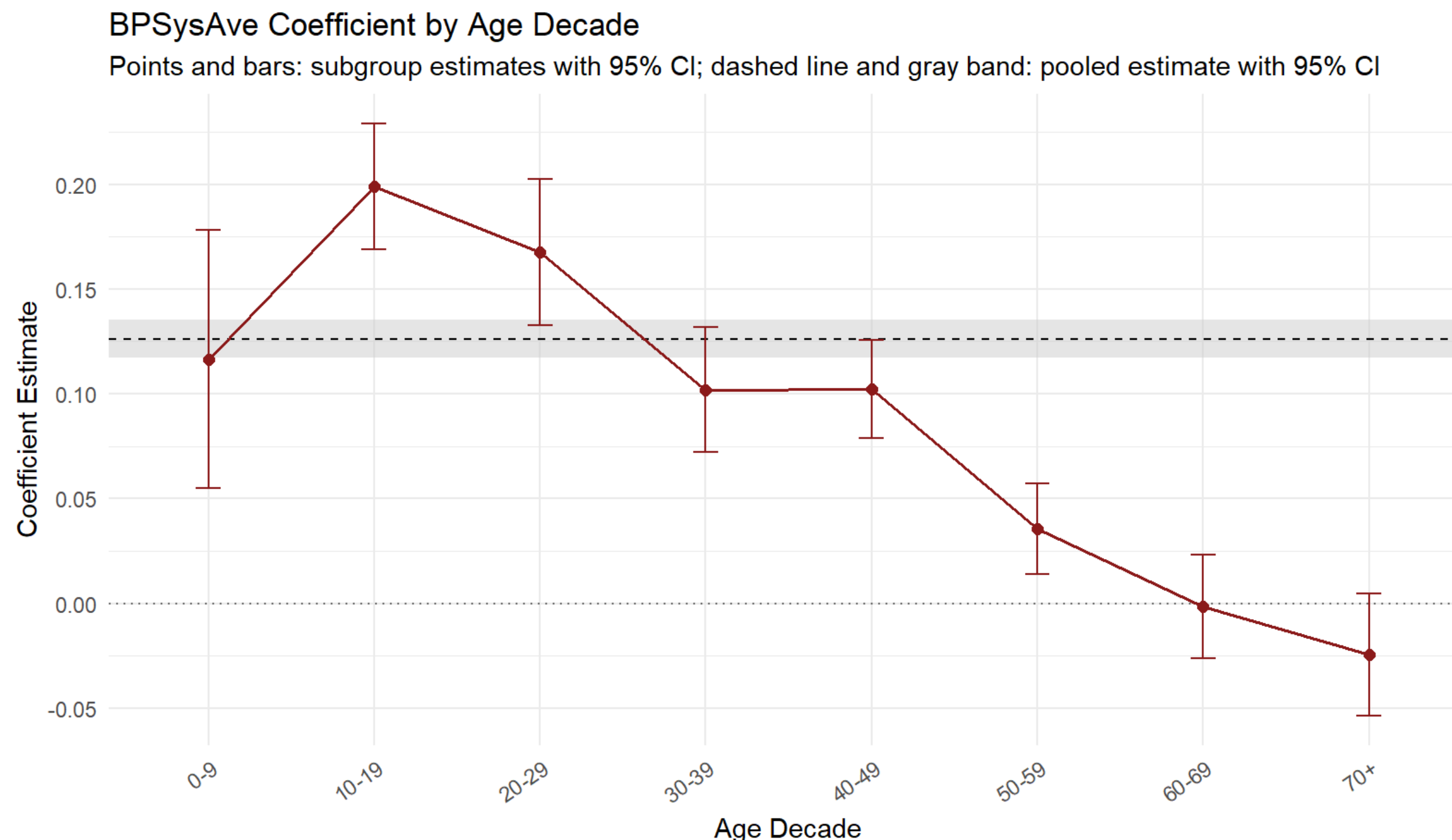
Simpson’s Paradox



Within each group the trend is positive. Pooled, it reverses. The pooled regression hides subgroup structure.

Real-Data: NHANES BPSysAve

NHANES 2017–2018 ($n = 9,254$). Response: systolic BP. Subgroups: age decades. Focus predictor: BPSysAve.



Metric	Value
SSR (BPSysAve)	0.57
Std. drift	32.07
Context-norm. drift	7.49

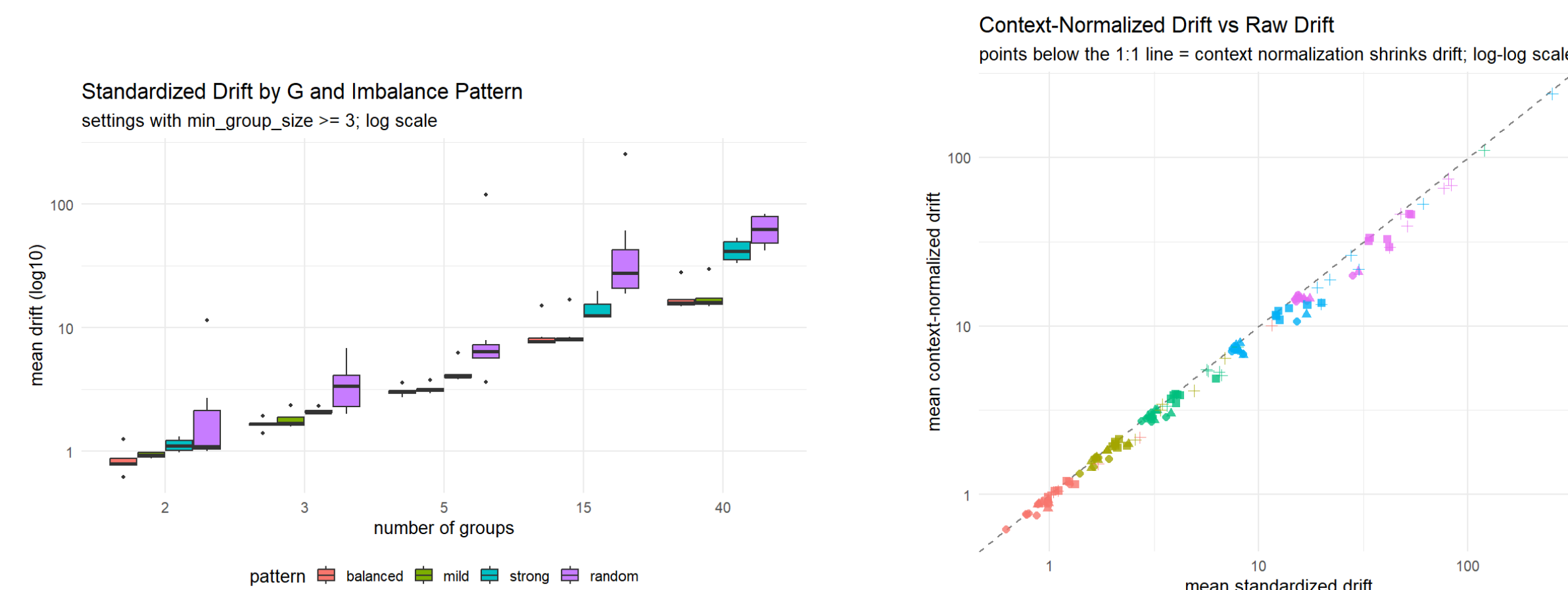
Key finding: Effect *reverses sign* in older groups. Drift $32 \rightarrow 7.5$ after context normalization, confirming genuine heterogeneity.

Four Models (nested)

- M1: pooled $Y = \beta_0 + X^\top \beta + \varepsilon$
- M2: add group intercept γG
- M3: add interactions $X^\top \delta G$
- M4: fully separate model per group

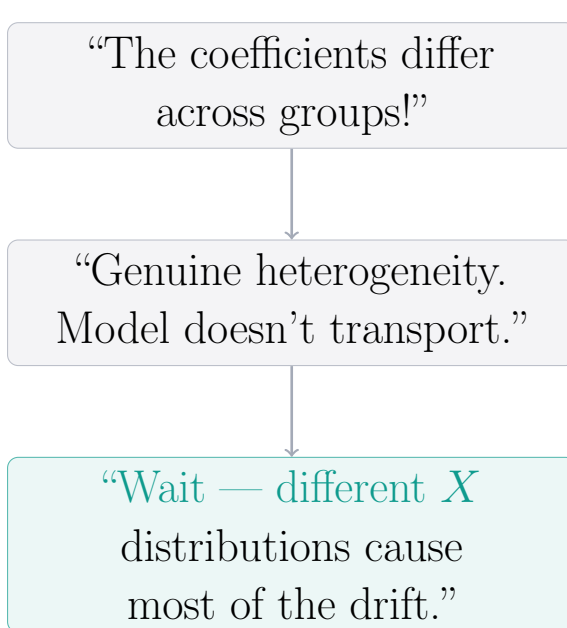
Simulation Grid (Scenario 1)

140 settings: $G \in \{3, 5, 10, 20, 40\}$, $n \in \{100, \dots, 30000\}$, 4 imbalance patterns (balanced, mild, strong, Dirichlet-random). 200 reps each.



Drift grows with G and imbalance (left). Context normalization shrinks drift (right).

“It’s an Artifact”



Coefficient differences come from **two sources**: genuine heterogeneity and mechanical artifact from covariate distribution differences. Standard practice conflates them.

Five Stability Metrics

Metric	Definition	What it captures
SSR	$SSR_j = \frac{1}{M} \sum \mathbf{1}(\text{sgn } \hat{\beta}_{j,a} = \text{sgn } \hat{\beta}_{j,b})$	Do conclusions (+/–) reverse across groups?
Standardized drift	$D_j^* = \frac{\max_g \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} }{\widehat{SE}(\hat{\beta}_{j,\text{pooled}})}$	How far do subgroup coefficients move from pooled, in SE units?
Context-norm. drift	$\tilde{D}_j = \frac{D_j^*}{1 + M_j}$, $M_j = \frac{1}{\binom{G}{2}} \sum_{a < b} \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	Drift after accounting for covariate distribution differences.
TPG	$TPG_{a \rightarrow b} = \text{RMSE}_{\text{pooled} \rightarrow b} - \text{RMSE}_{b \rightarrow b}$	Prediction loss when a model travels across groups.
CSS	$CSS_{a,b} = \frac{1}{p} \sum_j \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	How different are X distributions between groups?

What Drives Instability?

- Drift driven by G and per-group n , not total N .
- Strong/random imbalance amplifies drift 2–4× over balanced.
- SSR = 1.0 throughout (Scenario 1: identical β , no sign flips).
- Context normalization reduces drift; shrinkage $\propto$ CSS.

Future Work

Same analysis for Scenarios 2–4, then:

- Introduce **collinearity** across predictors
- Scale to **higher dimensions** (more predictors)
- Test **different regression models** beyond OLS

References

Berk et al. (2019); Tsoi et al. (2026); Bortsimas & Paskov (2020); Liu et al. (2020); Watson & Engle (1985); Garbade (1977); Buja et al. (2006).

github.com/RadJoubili/Regression-conclusion-stability-paper

Key Decomposition: Why Context Drift < Raw Drift

Even with identical β , different X distributions inflate drift:

$$\hat{\beta}^{(g)} - \hat{\beta}^{(\text{pooled})} = (X_g' X_g)^{-1} X_g' \varepsilon_g - (X' X)^{-1} X' \varepsilon$$

$$D_j^* = \tilde{D}_j \times (1 + M_j)$$

Total = Genuine × Mechanical

3 regimes: $M_j \approx 0 \Rightarrow$ all drift real; large M_j + small $\tilde{D}_j \Rightarrow$ artifact; both large $\Rightarrow$ genuine heterogeneity.